

HTML, CSS and SQL

1. HTML

HTML – HyperText Markup Language

- ❖ HTML is the standard markup language used to create and structure web pages. It tells the web browser what content should appear on a page and how that content is organized.
- ❖ HTML is not a programming language because it does not contain programming logic such as conditions and loops. It uses elements and tags to describe the structure of content.

History of HTML

- ❖ HTML was developed from the work of Tim Berners-Lee, who created the World Wide Web while working at CERN. The early HTML system was introduced around 1991. HTML later developed through several versions, including HTML 2.0, HTML 3.2, HTML 4.01 and HTML5. Modern HTML is maintained as a continuously updated Living Standard.

Web Browser

A web browser is a software application that allows users to access and view web pages on the Internet.

When we enter a website address, the browser requests the required files from a web server and displays the resulting web page.

Examples of Web Browsers

- ❖ Google Chrome
- ❖ Mozilla Firefox
- ❖ Microsoft Edge
- ❖ Safari
- ❖ Opera

How a Browser Works

❖ User → Browser → Web Server → HTML/CSS/JavaScript → Web Browser → Web Page

For example, when a user enters a website address, the browser sends a request to the server. The server sends the required files back. The browser interprets those files and displays the page.

Web Page

A web page is a document that can be displayed in a web browser.

A web page can contain:

- ❖ Text
- ❖ Images
- ❖ Videos
- ❖ Links
- ❖ Tables
- ❖ Forms
- ❖ Buttons
- ❖ Audio
- ❖ Animations

Web pages are commonly created using three major technologies:

HTML → Structure

CSS → Design

JavaScript → Behaviour

Example

A student registration page may contain:

- ❖ Student name
- ❖ Age
- ❖ Email
- ❖ Course
- ❖ Submit button
- ❖ HTML creates these elements, CSS makes them attractive, and JavaScript can provide interactive behaviour.

Simple HTML Program

HTML

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>My First Web Page</title>
```

```
</head>
```

```
<body>
```

```
  <h1>Welcome to My Website</h1>
```

```
  <p>This is my first HTML web page.</p>
```

```
</body>
```

```
</html>
```

Explanation

<!DOCTYPE html> Tells the browser that the document uses modern HTML.

```
<html>
```

The root element of the HTML document.

`<head>`

Contains information about the document.

`<title>`

Defines the title shown in the browser tab.

`<body>`

Contains the visible content of the web page.

`<h1>`

Creates a main heading.

`<p>`

Creates a paragraph.

Important HTML Tags

Tag

Purpose

`<html>`

Defines the HTML document

`<head>`

Contains document information

`<title>`

Sets the page title

`<body>`

Contains visible content

`<h1>`

Main heading

`<p>`

Paragraph

`<a>`

Creates a hyperlink

`<img>`

Displays an image

`<ul>`

Unordered list

`<ol>`

Ordered list

`<li>`

List item

<table>

Creates a table

<form>

Creates a form

<input>

Creates an input field

<button>

Creates a button

2. CSS

CSS – Cascading Style Sheets

CSS is a stylesheet language used to style and design HTML web pages.

HTML provides the structure, while CSS controls how that structure looks.

CSS is used for:

- ❖ Text color
- ❖ Font size
- ❖ Font style
- ❖ Background
- ❖ Borders
- ❖ Margins
- ❖ Padding
- ❖ Width and height

- ❖ Page layout
- ❖ Responsive design
- ❖ Animations

Types of CSS

There are three main types of CSS.

1. Inline CSS

CSS is written directly inside an HTML element.

HTML

```
<p style="color: blue;">Welcome</p>
```

2. Internal CSS

CSS is written inside the <style> element.

HTML

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <style>
```

```
    h1 {
```

```
      color: blue;
```

```
    }
```

```
p {  
    font-size: 20px;  
}  
  
</style>  
</head>  
  
<body>  
    <h1>Welcome</h1>  
    <p>Hello World</p>  
  
</body>  
  
</html>
```

3. External CSS

CSS is written in a separate .css file.

HTML:

HTML

```
<link rel="stylesheet" href="style.css">
```

style.css:

```
h1 {  
    color: blue;
```



```
}
```

```
p {  
    font-size: 20px;  
}
```

External CSS is commonly preferred for larger websites because one stylesheet can be used across multiple pages.

CSS Basic Syntax

```
selector {  
    property: value;  
}
```

Example:

```
h1 {  
    color: blue;  
    font-size: 30px;  
}
```

Here:

h1 = Selector

color = Property

blue = Value

font-size = Property

30px = Value

3. SQL

Full Form

SQL – Structured Query Language

SQL is a language used to communicate with and manage relational databases.

A database stores organized information.

For example, a student database may contain:

ID

Name

Course

1

Riswana B

Java

2

Safrosh B

HTML

3

Afrin B

SQL

SQL allows us to create tables and perform operations on this data.

Main SQL Operations

SQL commonly performs CRUD operations:

C – Create

Add new data.

R – Read

Retrieve existing data.

U – Update

Modify existing data.

D – Delete

Remove data.

Simple SQL Example

Create a Table

CREATE TABLE Students (

id INT,

```
name VARCHAR(50),  
age INT  
);
```

This creates a table called Students.

Insert Data

```
INSERT INTO Students  
VALUES (1, 'Riswana B', 20);
```

This adds a student record.

Read Data

```
SELECT * FROM Students;
```

This displays all records from the table.

Update Data

```
UPDATE Students
```

```
SET age = 23
```

```
WHERE id = 1;
```

This changes the age of the student whose ID is 1.

Delete Data

```
DELETE FROM Students
```

WHERE id = 1;

This removes that particular record.

Important SQL Commands

Command

Purpose

CREATE

Creates database objects

INSERT

Adds data

SELECT

Retrieves data

UPDATE

Modifies data

DELETE

Removes data

ALTER

Changes table structure

DROP

Removes a database object

WHERE

Filters records

ORDER BY

Sorts results

GROUP BY

Groups records

HTML vs CSS vs SQL

Technology

Main Purpose

HTML

Creates web page structure

CSS

Designs and styles the web page

SQL

Stores and manages database information

Simple Example

For an online student registration website:

HTML creates:

Name box, Email box, Course box and Submit button.

CSS controls:

Colors, fonts, spacing, layout and appearance.

SQL stores:

Student name, email, course and other database information.

Final Concept

HTML → Structure

CSS → Style

JavaScript → Function/Interaction

SQL → Database